

Ndumiso Baliti

Exercise 9

$$i) \frac{\partial^2 u}{\partial x^2} = \frac{u(x + \Delta x) - 2u(x) + u(x - \Delta x)}{\Delta x^2}$$

grid: $x_i = x_0 + i \Delta x$
 $t^n = t_0 + n \Delta t$

$$u(x_i, t^n) = U_i^n \quad ; \quad u(x_i - \Delta x, t^n) = U_{i-1}^n$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{U_{i+1}^n - 2U_i^n + U_{i-1}^n}{\Delta x^2}$$

$$ii) \frac{\partial u}{\partial z} = \frac{u(z + \Delta z) - u(z - \Delta z)}{2\Delta z}$$

grid: $z_k = z_0 + k \Delta z$

$$u(x_i, y_j, z_k, t^n) = U_{ij,k}^n \quad ; \quad u(x_i, y_j, z_k + \overset{\text{(above)}}{\Delta z}, t^n) = U_{ij,(k+1)}^n$$

$$u(x_i, y_j, z_k - \overset{\text{(below)}}{\Delta z}, t^n) = U_{ij,(k-1)}^n$$

$$\frac{\partial u}{\partial z} = \frac{U_{ij,(k+1)}^n - U_{ij,(k-1)}^n}{2\Delta z}$$